From Gaussians to a Cluster Mass Map

A worksheet on likelihood stacking, reference shears and κ reconstruction — Answer key

KL-NN paper, Sec. 6.4–6.5 • October 2026

How to use this worksheet

- The problems build on each other. Each one has a **Given** box with everything you need that is not a result of an earlier problem. A "You'll need from before" line names the earlier results it uses.
- Every Given box names its **source**: a paper, section and equation number, read in the source text. A result that we derived ourselves and that is not in those sources is marked *derived here*. The full list of sources is at the end, and `references.md` quotes the line each citation points to.
- Problems marked [Py] have a companion cell in `cluster_worksheet.ipynb` (numpy, scipy, matplotlib). Do the pen-and-paper part first, then check it numerically.
- Parts A–B are statistics, C–D are lensing and Fourier analysis, E–F are regularised inversion, and G is the cluster analysis itself, assembled from the earlier parts.

Notation

$\mathbf{g} = (g_1, g_2)$	reduced shear of a galaxy; γ shear, κ convergence
$L_i(\mathbf{g})$	shear likelihood of galaxy i (its NPE posterior divided by the training prior)
C_i, $F_i = C_i^{-1}$	covariance and precision (inverse covariance) of L_i
R_i	rotation from the image frame to galaxy i's tangential/cross frame
$\mathbf{G}$, $\hat{\mathbf{G}}$	common shear of a cell; its stacked estimate (posterior mean)
$\mathbf{g}_{\mathrm{ref}}$	reference shear of a cell (Problem B4)
m, c	multiplicative and additive bias, $\hat{g} = (1+m)g + c$
H, H_d, H_p	Hessian of the negative log posterior; its data and prior parts
R	resolution matrix, $\mathbb{E}[\hat{\kappa}] = \mathsf{R}\,\kappa_{\mathrm{true}}$

Contents

Part A. Combining measurements **2**

A1 Two measurements of one number . . . 2
A2 Many measurements . . . 2
A3 Two-component measurements with covariance matrices . . . 3
A4 Posterior, prior and likelihood . . . 3
A5 [Py] Why one galaxy's posterior mean is shrunk . . . 4

Part B. Estimators and stacking **5**

B1 Bias, scatter and pulls . . . 5
B2 Multiplicative and additive bias; shrinkage of a posterior mean . . . 5
B3 Stack likelihoods, not posteriors . . . 6
B4 Fitting one shear to galaxies whose shears differ: the reference shear . . . 6
B5 [Py] The tangential frame, and checking a stack with pulls . . . 7

Part C. Weak-lensing basics **8**
C1 One potential gives both mass and shear . . . 8
C2 Tangential and cross shear . . . 8
C3 Reduced shear and the mass-sheet degeneracy . . . 9
C4 Adding two halos . . . 9
C5 Where the subhalo's signal is, and how strong it is . . . 10

Part D. Fourier space and Kaiser–Squires **10**
D1 Derivatives become multiplications . . . 10
D2 The Kaiser–Squires kernel . . . 11
D3 Finite fields, padding and the mean mass . . . 11
D4 [Py] What binning does to a tangential pattern . . . 12

Part E. Regularised inversion and the Wiener filter **13**
E1 When the data can't determine the answer . . . 13
E2 A Gaussian prior makes the problem well posed . . . 13
E3 One mode at a time: what the Wiener filter keeps . . . 14
E4 A smoothness penalty is a power spectrum . . . 14
E5 [Py] A one-dimensional Wiener filter . . . 15

Part F. Uncertainty, resolution and effective parameters **15**
F1 The Laplace approximation . . . 15
F2 Gauss–Newton: curvature from the model, not from the data grid . . . 16
F3 Two different uncertainties, and the resolution matrix . . . 16
F4 [Py] How many numbers do the data actually determine? . . . 17

Part G. The cluster analysis **18**
G1 Bernstein & Armstrong for a constant shear . . . 18
G2 Bernstein & Armstrong for a varying shear . . . 18
G3 Why the difference map is clean . . . 19
G4 [Py] The significance one map can claim . . . 20
G5 Reading the ensemble numbers . . . 20
G6 Why the map settings were chosen on synthetic data . . . 21

Sources **21**

Part A. Combining measurements

A1 Two measurements of one number

Given. A measurement d of an unknown x with Gaussian noise of standard deviation σ has likelihood

$$L(x) \propto \exp\left[-\frac{(x-d)^2}{2\sigma^2}\right].$$

Independent measurements multiply: the joint likelihood of d_1 and d_2 is $L_1(x)\,L_2(x)$. Maximising it is the same as minimising $\chi^2 = \sum_i (d_i - x)^2/\sigma_i^2$, the weighted least-squares problem.

Source: HBL10, Sec. 1 (Standard practice), Eqs. (5) and (7), with the parameter covariance $[A^\mathsf{T} C^{-1} A]^{-1}$ stated in the text after Eq. (7).

You'll need from before: nothing; this is the start.

(a) Show that $L_1(x)L_2(x)$ is again a Gaussian in x. Find its mean $\hat{x}$ and variance $\sigma_{\hat{x}}^2$.

(b) Write $\hat{x}$ as a weighted average of d_1 and d_2. What are the weights?

(c) A KL galaxy measures one shear component as $d_1 = 0.03 \pm 0.05$; a second galaxy gives $d_2 = 0.01 \pm 0.02$. Compute $\hat{x}$ and $\sigma_{\hat{x}}$. Which galaxy dominates, and why?

Solution. (a) Adding the exponents, $-\frac{1}{2}\left[(x-d_1)^2/\sigma_1^2 + (x-d_2)^2/\sigma_2^2\right] = -\frac{1}{2}(w_1+w_2)\,x^2 + (w_1 d_1 + w_2 d_2)\,x + \text{const}$, with $w_i = 1/\sigma_i^2$. Completing the square gives a Gaussian with

$$\hat{x} = \frac{w_1 d_1 + w_2 d_2}{w_1 + w_2}, \qquad \sigma_{\hat{x}}^2 = \frac{1}{w_1 + w_2}.$$

(b) The weights are the precisions $w_i = 1/\sigma_i^2$, normalised to sum to 1. Precisions add. (c) $w_1 = 400$, $w_2 = 2500$: $\hat{x} = (12 + 25)/2900 = 0.0128$ and $\sigma_{\hat{x}} = 2900^{-1/2} = 0.0186$. The second galaxy dominates: its weight is $(0.05/0.02)^2 = 6.25$ times larger. The combined error is smaller than either.

A2 Many measurements

Given. The result of A1 generalises to N independent Gaussian measurements $d_i \pm \sigma_i$ of one x: the product of the likelihoods is Gaussian, with precision $\sum_i w_i$ and mean $\sum_i w_i d_i / \sum_i w_i$, where $w_i = 1/\sigma_i^2$.

Source: HBL10, Sec. 1 (Standard practice), Eqs. (5) and (7), with the parameter covariance $[A^\mathsf{T} C^{-1} A]^{-1}$ stated in the text after Eq. (7).

You'll need from before: A1 (precisions add; the mean is precision-weighted).

(a) Show that for equal errors $\sigma_i = \sigma$ the combined error is $\sigma/\sqrt{N}$.

(b) A KL galaxy measures each shear component with $\sigma_g \approx 0.05$. How many such galaxies are needed for an error of 0.01? What is the error for a cell of 5 galaxies?

(c) Our cluster cells hold about 5 galaxies. The subhalo changes the shear in most cells by $\lesssim 0.01$. Can a single cell detect the subhalo?

Solution. (a) $\sum_i w_i = N/\sigma^2$, so $\sigma_{\hat{x}} = \sigma/\sqrt{N}$. (b) $0.05/\sqrt{N} = 0.01$ gives $N = 25$. With 5 galaxies, $0.05/\sqrt{5} = 0.022$. This is the per-cell error seen in the cluster shear maps. (c) No: a $\lesssim 0.01$ signal against a 0.022 error is below 0.5σ per cell. Detecting the subhalo needs the information of many cells (or many galaxies) combined. That is what the κ map does (Parts E–G).

A3 Two-component measurements with covariance matrices

Given. A measurement $\mathbf{d}_i$ of a 2-vector $\mathbf{x}$ (for us, $\mathbf{x} = (g_1, g_2)$) with Gaussian noise of covariance C_i has likelihood $L_i(\mathbf{x}) \propto \exp\big[-\tfrac{1}{2}(\mathbf{x}-\mathbf{d}_i)^\mathsf{T} F_i\,(\mathbf{x}-\mathbf{d}_i)\big]$, with precision matrix $F_i = C_i^{-1}$. The general weighted least-squares solution, for data Y with covariance C and a linear model $Y = AX$, is $\hat{X} = [A^\mathsf{T} C^{-1} A]^{-1} A^\mathsf{T} C^{-1} Y$, with covariance $[A^\mathsf{T} C^{-1} A]^{-1}$.

Source: HBL10, Sec. 1 (Standard practice), Eqs. (5) and (7), with the parameter covariance $[A^\mathsf{T} C^{-1} A]^{-1}$ stated in the text after Eq. (7).

You'll need from before: A1, A2.

(a) Show that $\prod_i L_i(\mathbf{x})$ is Gaussian with precision $F_{\rm tot} = \sum_i F_i$ and mean $\hat{\mathbf{x}} = F_{\rm tot}^{-1} \sum_i F_i \mathbf{d}_i$. (Either complete the square, or use the weighted least-squares solution with A a stack of 2×2 identity matrices.)

(b) Two galaxies: $\mathbf{d}_1 = (0.02, 0)$ with $F_1 = \mathrm{diag}(2500, 100)$; $\mathbf{d}_2 = (0, 0.02)$ with $F_2 = \mathrm{diag}(100, 2500)$. Compute $\hat{\mathbf{x}}$. Compare with a "scalar" average that weights each galaxy by the trace of its F_i.

(c) Now $\mathbf{d}_1 = (0.02, 0)$ with $F_1 = \begin{pmatrix} 1300 & 1200 \\ 1200 & 1300 \end{pmatrix}$ and $\mathbf{d}_2 = (0, 0)$ with $F_2 = 100\,I$. Compute $\hat{\mathbf{x}}$. Neither measurement has a non-zero second component; why is $\hat{x}_2 \neq 0$?

(d) KL kinematics measure one shear direction much better than the other. What does this problem say about combining KL galaxies?

Solution. (a) $\sum_i (\mathbf{x}-\mathbf{d}_i)^\mathsf{T} F_i (\mathbf{x}-\mathbf{d}_i) = \mathbf{x}^\mathsf{T} F_{\rm tot} \mathbf{x} - 2\mathbf{x}^\mathsf{T} \sum_i F_i \mathbf{d}_i + \text{const} = (\mathbf{x}-\hat{\mathbf{x}})^\mathsf{T} F_{\rm tot} (\mathbf{x}-\hat{\mathbf{x}}) + \text{const}$, with $\hat{\mathbf{x}} = F_{\rm tot}^{-1} \sum_i F_i \mathbf{d}_i$. (b) $F_{\rm tot} = \mathrm{diag}(2600, 2600)$ and $\sum F_i \mathbf{d}_i = (50, 50)$, so $\hat{\mathbf{x}} = (0.0192, 0.0192)$. Each galaxy dominates the component it measures well. The trace weights are equal (2600 each), giving $(0.01, 0.01)$: half the signal is lost, because the scalar average throws away which direction each galaxy measures. (c) $F_{\rm tot} = \begin{pmatrix} 1400 & 1200 \\ 1200 & 1400 \end{pmatrix}$ and $\sum F_i \mathbf{d}_i = (26, 24)$. Then $F_{\rm tot}^{-1} = \frac{1}{520000} \begin{pmatrix} 1400 & -1200 \\ -1200 & 1400 \end{pmatrix}$, so $\hat{\mathbf{x}} = (0.0146, 0.0046)$. F_1 has off-diagonal terms: galaxy 1 measures the combination $x_1 + x_2$ well (eigenvalue 2500 along $(1,1)/\sqrt{2}$) and $x_1 - x_2$ poorly (eigenvalue 100). The components mix. (d) Matrix weights are required; scalar weights mislabel the result. The same matrix weighting reappears in the reference shear (B4) and in BA14 (G1).

A4 Posterior, prior and likelihood

Given. Bayes' rule: $p(x \mid d) = L(x)\,\pi(x)/Z$, with $L(x) = p(d \mid x)$ the likelihood, $\pi(x)$ the prior and Z a normalising constant. A neural posterior estimator (NPE) trained on simulations whose

parameters were drawn from $\pi_{\rm train}$ learns $q(x \mid d) \approx p(x \mid d)$ under that prior. If posterior samples were drawn under an *interim* prior, dividing by that prior (reweighting the samples) recovers the likelihood.

Source: H12, Sec. 1 (Generalities), Eqs. (6) and (9)–(11), and Sec. 3 (Posterior probabilities), Eqs. (37)–(38); Sch15, Sec. 2.2.2 (Hierarchical inference via importance sampling), Eqs. (13)–(15); HMB10, Sec. 2 (Method), Eqs. (5)–(9).

You'll need from before: nothing beyond the definitions.

(a) Show that $q(x \mid d)/\pi_{\rm train}(x) \propto L(x)$ wherever $\pi_{\rm train}(x) > 0$.

(b) Our training prior is uniform on the box $|g_k| < 0.1$. What does $q/\pi_{\rm train}$ look like inside the box, and what do we know about L outside it?

(c) Why must we divide by the prior the network was *trained* with, not by some other prior we prefer?

Solution. (a) $q = L\,\pi_{\rm train}/Z$, so $q/\pi_{\rm train} = L/Z \propto L$ (constant in x). (b) Inside the box $\pi_{\rm train}$ is constant, so $q/\pi_{\rm train} \propto q$: the likelihood has the posterior's shape there. Outside the box $q = 0$, so the likelihood is unknown there: the NPE says nothing about shears it never saw. (c) The network has absorbed $\pi_{\rm train}$ into q. Dividing by a different π' leaves a factor $\pi_{\rm train}/\pi'$ that is not part of the likelihood, which would bias every later product of likelihoods.

A5 [Py] Why one galaxy's posterior mean is shrunk

Given. Take one shear component. The likelihood is Gaussian, $L(x) \propto \exp[-(x-d)^2/2\sigma^2]$, with $\sigma = 0.07$ (the untruncated width of a typical KL likelihood), and the prior is uniform on $|x| < 0.1$. The posterior is then a Gaussian truncated to $[-0.1, 0.1]$. Its mean is the "posterior-mean shear estimate" of one galaxy. The data scatter as $d \sim \mathcal{N}(x_{\rm true}, \sigma^2)$. *Derived here.*

You'll need from before: A4 (posterior = likelihood × prior).

(a) Let $\alpha = (-0.1 - d)/\sigma$, $\beta = (0.1 - d)/\sigma$, and let ϕ and Φ be the standard normal density and CDF. Show that the posterior mean is $d + \sigma\,[\phi(\alpha) - \phi(\beta)]/[\Phi(\beta) - \Phi(\alpha)]$. (Integrate $x\,e^{-(x-d)^2/2\sigma^2}$ by substitution.)

(b) Explain without calculation why its average over d is smaller in magnitude than $x_{\rm true}$.

(c) [Py] Compute $\mathbb{E}_d$[posterior mean] for $x_{\rm true} \in [-0.05, 0.05]$, fit a line $(1+m)x_{\rm true} + c$, and report m.

(d) [Py] Now average N independent galaxies at the same $x_{\rm true}$ by multiplying their likelihoods, which is a Gaussian of width $\sigma/\sqrt{N}$ by A2, truncated to the box. Repeat (c) for $N = 1, 2, 4, 8, 16$. How large must N be for $|m| < 0.01$?

Solution. (a) With $u = (x-d)/\sigma$, $\int_{-0.1}^{0.1} x\,\phi(u)\,dx \propto d\,[\Phi(\beta) - \Phi(\alpha)] + \sigma \int_\alpha^\beta u\,\phi(u)\,du$, and $\int_\alpha^\beta u\phi\,du = \phi(\alpha) - \phi(\beta)$. Dividing by the normalisation $\Phi(\beta) - \Phi(\alpha)$ gives the result. (b) When d falls near or beyond an edge, the box cuts off the far tail of the likelihood, so the mean is

dragged inward. For $x_{\rm true} > 0$ the upper edge is reached more often than the lower one, so on average the estimate is pulled toward zero. (c) $m \approx -0.57$ for $\sigma = 0.07$. A single posterior mean is strongly shrunk, close to the $m \approx -0.6$ of the raw NPE posterior means (paper Sec. 6.1). (d) $m \approx -0.57, -0.36, -0.17, -0.052, -0.0085$ for $N = 1, 2, 4, 8, 16$ (and -0.0005 for $N = 32$). The product narrows as $\sigma/\sqrt{N}$ and stops touching the box, so the shrinkage disappears. With every galaxy at $\sigma = 0.07$, $|m| < 0.01$ needs $N \approx 16$. Real KL widths vary from galaxy to galaxy, and the narrow ones dominate a product, which is why the paper's stacks reach percent-level m at smaller N. Multiplying likelihoods, rather than averaging posterior means, is what removes the bias.

Part B. Estimators and stacking

B1 Bias, scatter and pulls

Given. An estimator $\hat{\theta}$ of a true value θ comes with a reported error bar $\hat{\sigma}$. The *pull* is $(\hat{\theta} - \theta)/\hat{\sigma}$. Suppose $\hat{\theta} \sim \mathcal{N}(\theta + b, s^2)$: bias b, true scatter s. *Derived here*; the pull is a standard diagnostic.

You'll need from before: nothing beyond Gaussians.

(a) With $\hat{\sigma}$ fixed, show that the pull is $\mathcal{N}(b/\hat{\sigma},\ s^2/\hat{\sigma}^2)$.

(b) What do "pull mean $\neq 0$" and "pull spread $\neq 1$" each tell you?

(c) Our 50-realization ensemble gives a pull spread of 0.84. By what factor do the reported error bars exceed the true scatter? If you used them in an inference, what fraction of the available information ($\propto 1/\sigma^2$) would you be claiming?

(d) The polar cells give 1437 pulls. What is the statistical uncertainty on their mean? Is a pull mean of -0.02 evidence of bias?

Solution. (a) $(\hat{\theta} - \theta)/\hat{\sigma}$ is a linear function of a Gaussian, with mean $b/\hat{\sigma}$ and standard deviation $s/\hat{\sigma}$. (b) A non-zero mean means bias, in units of the error bar. A spread above 1 means the error bars are too small (overconfident); below 1, too large (conservative). (c) $\hat{\sigma}/s = 1/0.84 = 1.19$, so the errors are 19% too large. You would claim $s^2/\hat{\sigma}^2 = 0.84^2 = 0.71$ of the information, as if you had thrown away 29% of the galaxies. This loses precision but introduces no bias. (d) $0.84/\sqrt{1437} = 0.022$. A mean of -0.02 is 0.9σ: consistent with zero.

B2 Multiplicative and additive bias; shrinkage of a posterior mean

Given. Shear biases are summarised by a linear response $\hat{g} = (1+m)\, g + c$, with m the multiplicative and c the additive bias

Source: BA14, Sec. 1 (Introduction), Eq. (1). Model one galaxy with a Gaussian likelihood $d \sim \mathcal{N}(g, \sigma^2)$ and a Gaussian prior $g \sim \mathcal{N}(0, \sigma_\pi^2)$. *Derived here* (it reproduces the toy model of paper Sec. 6.2).

You'll need from before: A1 (product of two Gaussians).

(a) Treat the prior as a second "measurement" $0 \pm \sigma_\pi$. Show that the posterior mean is $\hat{g} = d\, \sigma_\pi^2/(\sigma_\pi^2 + \sigma^2)$.

(b) Average over the noise to find $\mathbb{E}[\hat{g}]$ and read off m and c.

(c) Use $\sigma = 0.07$ and the standard deviation of a uniform prior on $[-0.1, 0.1]$, $\sigma_\pi = 0.2/\sqrt{12}$, to estimate m. Compare with A5(c).

Solution. (a) By A1, with weights $1/\sigma^2$ on d and $1/\sigma_\pi^2$ on 0: $\hat{g} = (d/\sigma^2)/(1/\sigma^2 + 1/\sigma_\pi^2) = d\,\sigma_\pi^2/(\sigma_\pi^2 + \sigma^2)$. (b) $\mathbb{E}[d] = g$, so $\mathbb{E}[\hat{g}] = g\,\sigma_\pi^2/(\sigma_\pi^2 + \sigma^2)$: $m = -\sigma^2/(\sigma^2 + \sigma_\pi^2)$ and $c = 0$. (c) $\sigma_\pi = 0.0577$, so $m = -0.0049/(0.0049 + 0.00333) = -0.60$. The truncated box of A5 gave -0.57; the Gaussian toy captures the effect well. Shrinkage is strongest when the likelihood is wide compared with the prior.

B3 Stack likelihoods, not posteriors

Given. Each galaxy's NPE posterior is $q_i(g) \propto L_i(g)\,\pi_{\text{train}}(g)$ (A4). N galaxies share one g and have independent data. The correct joint posterior is $p(g \mid d_1, \ldots, d_N) \propto \pi(g) \prod_i L_i(g)$, and samples drawn under an interim prior are combined by dividing that prior out.

Source: BA14, Sec. 2.1 (General formulation), Eqs. (2)–(7); Sch15, Sec. 2.2.2 (Hierarchical inference via importance sampling), Eqs. (13)–(15); HMB10, Sec. 2 (Method), Eqs. (5)–(9).

You'll need from before: A4, B2.

(a) Show that $\prod_i q_i(g) \propto \pi_{\text{train}}(g)^N \prod_i L_i(g)$. How many times is the prior counted?

(b) Use the Gaussian model of B2 (prior $\mathcal{N}(0, \sigma_\pi^2)$, each likelihood width σ). Show that the mean of $\prod_i q_i$ is shrunk by the same factor $\sigma_\pi^2/(\sigma_\pi^2 + \sigma^2)$ for every N.

(c) What is the mean of the correct posterior $\pi \prod_i L_i$, and how does its m behave as $N \to \infty$?

(d) Why does our stack use $L_i = q_i/\pi_{\text{train}}$ and a single flat prior on the cell shear?

Solution. (a) $\prod_i q_i \propto \prod_i L_i \pi_{\text{train}} = \pi_{\text{train}}^N \prod_i L_i$: the prior is counted N times. (b) π_{train}^N has precision N/σ_π^2 and $\prod L_i$ has precision N/σ^2 about the mean $\bar{d}$, so the mean is $\bar{d}\,(N/\sigma^2)/(N/\sigma^2 + N/\sigma_\pi^2) = \bar{d}\,\sigma_\pi^2/(\sigma_\pi^2 + \sigma^2)$: the N cancels, and $m = -0.6$ no matter how many galaxies you multiply. (c) $\bar{d}\,(N/\sigma^2)/(N/\sigma^2 + 1/\sigma_\pi^2)$, so $m = -\sigma^2/(\sigma^2 + N\sigma_\pi^2) \to 0$ as $N \to \infty$. (d) Dividing out π_{train} leaves the likelihoods, which multiply correctly. The cell then has one prior (uniform on $|\mathbf{G}| < 0.1$), counted once, and the data dominate as N grows.

B4 Fitting one shear to galaxies whose shears differ: the reference shear

Given. Fitting a model that cannot be exactly right (a misspecified model) converges, as the data grow, to the parameter value that makes the model's distribution closest to the true one in Kullback–Leibler divergence: the *pseudo-true* value.

Source: KvdV12, Sec. 1 (Introduction) and Sec. 2.2 (Asymptotic normality in the i.i.d. case), Eq. (2.7). The original result is White (1982), which is not openly available and was not checked. In a cell, galaxy i has true shear $\mathbf{t}_i$ (in the cell frame), a Gaussian likelihood centred on $\mathbf{d}_i$ with precision F_i, and $\mathbb{E}[\mathbf{d}_i] = \mathbf{t}_i$. The stack fits one common $\mathbf{G}$ to all of them.

You'll need from before: A1, A3 (the matrix-weighted mean).

(a) By A3, $\hat{\mathbf{G}} = F_{\rm tot}^{-1}\sum_i F_i \mathbf{d}_i$. Average over the noise, assuming F_i doesn't depend on it. Show that $\mathbb{E}[\hat{\mathbf{G}}] = F_{\rm tot}^{-1}\sum_i F_i \mathbf{t}_i \equiv \mathbf{g}_{\rm ref}$.

(b) Show that $\mathrm{Cov}(\hat{\mathbf{G}}) = F_{\rm tot}^{-1}$, and that it doesn't depend on how much the $\mathbf{t}_i$ differ. Why is this still the right error bar *about* $\mathbf{g}_{\rm ref}$?

(c) Scalar example: galaxy 1 has $t_1 = 0.02$, $\sigma_1 = 0.02$; galaxy 2 has $t_2 = 0$, $\sigma_2 = 0.06$. Compute $g_{\rm ref}$ and the plain mean. Which one does the stack estimate?

(d) Why would comparing the stack with the shear at the cell centre be the wrong test?

Solution. (a) $\mathbb{E}[\hat{\mathbf{G}}] = F_{\rm tot}^{-1}\sum_i F_i\,\mathbb{E}[\mathbf{d}_i] = F_{\rm tot}^{-1}\sum_i F_i\mathbf{t}_i$. This is the pseudo-true value for Gaussian likelihoods: the constant that best describes the varying $\mathbf{t}_i$, with each galaxy weighted by its precision. (b) With independent noises of covariance F_i^{-1}: $\mathrm{Cov}(\hat{\mathbf{G}}) = F_{\rm tot}^{-1}\big(\sum_i F_i F_i^{-1} F_i\big)F_{\rm tot}^{-1} = F_{\rm tot}^{-1}$. The $\mathbf{t}_i$ don't appear. It is the right error about $\mathbf{g}_{\rm ref}$ because $\mathbf{g}_{\rm ref}$ is exactly what $\hat{\mathbf{G}}$ estimates; so $\hat{\mathbf{G}} - \mathbf{g}_{\rm ref} \sim \mathcal{N}(0, F_{\rm tot}^{-1})$, and the pulls about $\mathbf{g}_{\rm ref}$ are standard normal. (c) $w_1 = 2500$, $w_2 = 278$: $g_{\rm ref} = 50/2778 = 0.0180$; plain mean 0.01. The stack estimates 0.0180, with error 0.019. Comparing it with 0.01 would call a correct stack biased. (d) The cell-centre shear is not what the stack converges to. The difference is a deterministic binning effect, largest where the shear changes quickly. (Full derivation: `CELL_REFERENCE_SHEAR.md`, Secs. 3–5.)

B5 [Py] The tangential frame, and checking a stack with pulls

Given. About a lens at the origin, a galaxy at position angle ϕ has tangential and cross shear $g_t = -\mathrm{Re}(g\,e^{-2i\phi})$, $g_\times = -\mathrm{Im}(g\,e^{-2i\phi})$, with $g = g_1 + ig_2$.

Source: B&S01, Sec. 5.3.1 (Aperture Mass Measures), Eq. (5.25) for γ_t; Kilbinger15, Sec. 3.8 (The shear correlation function), Eq. (31) for γ_t and $\gamma_\times$. As a matrix, $(g_t, g_\times)^{\mathsf{T}} = R(\phi)\,(g_1, g_2)^{\mathsf{T}}$ with $R(\phi) = -\begin{pmatrix} \cos 2\phi & \sin 2\phi \\ -\sin 2\phi & \cos 2\phi \end{pmatrix}$.

You'll need from before: A3, B1, B4.

(a) Show that $R(\phi)$ is orthogonal ($RR^{\mathsf{T}} = I$), and that a likelihood with precision F in the image frame has precision $F' = RFR^{\mathsf{T}}$ in the tangential frame.

(b) An axisymmetric lens produces purely tangential shear: $g_\times = 0$ and $g_t > 0$. Write (g_1, g_2) at angle ϕ in terms of g_t.

(c) [Py] Simulate a cell of 5 galaxies at different ϕ around a lens with g_t varying from 0.01 to 0.03 and anisotropic, randomly oriented likelihood covariances. Draw noisy measurements, rotate each into the tangential frame, stack with A3, and repeat 2000 times. Show that the pulls about $\mathbf{g}_{\rm ref}$ have mean ≈ 0 and spread ≈ 1, while the pulls about the plain mean of the $g_{t,i}$ are biased.

Solution. (a) Writing $R = -Q$ with Q a rotation by 2ϕ, $RR^{\mathsf{T}} = QQ^{\mathsf{T}} = I$. If $\ln L \propto -\frac{1}{2}(\mathbf{g} - \hat{\mathbf{g}})^{\mathsf{T}}F(\mathbf{g} - \hat{\mathbf{g}})$ and $\mathbf{g} = R^{\mathsf{T}}\mathbf{g}'$, then $(\mathbf{g} - \hat{\mathbf{g}})^{\mathsf{T}}F(\mathbf{g} - \hat{\mathbf{g}}) = (\mathbf{g}' - R\hat{\mathbf{g}})^{\mathsf{T}}RFR^{\mathsf{T}}(\mathbf{g}' - R\hat{\mathbf{g}})$, so $F' = RFR^{\mathsf{T}}$. (b) $g\,e^{-2i\phi} = -g_t$ gives $g = -g_t e^{2i\phi}$: $g_1 = -g_t\cos 2\phi$, $g_2 = -g_t \sin 2\phi$. (c) See the solutions notebook: the pulls about $\mathbf{g}_{\rm ref}$ have mean and spread close to 0 and 1. The pulls about the plain mean are offset (by about 0.05–0.1σ in this example; more when the precisions differ more), because precise galaxies at particular g_t dominate the stack.

Part C. Weak-lensing basics

C1 One potential gives both mass and shear

Given. A thin lens is described by the lensing potential $\psi(\boldsymbol{\theta})$. Its second derivatives ($\psi_{ij} = \partial^2\psi/\partial\theta_i\partial\theta_j$) give the convergence (projected mass in units of the critical density) and the two shear components:

$$\kappa = \tfrac{1}{2}(\psi_{11} + \psi_{22}), \qquad \gamma_1 = \tfrac{1}{2}(\psi_{11} - \psi_{22}), \qquad \gamma_2 = \psi_{12}.$$

Source: B&S01, Sec. 3.1.3 (Magnification and Distortion), Eq. (3.12); Kilbinger15, Sec. 3.3 (Linearized lensing quantities), Eq. (19). A singular isothermal sphere (SIS) has $\psi = \theta_E\,|\boldsymbol{\theta}|$, with Einstein radius θ_E.

Source: B&S01, Sec. 3.1.5 (An Illustrative Example: Isothermal Spheres), Eqs. (3.17)–(3.18), with $\psi = \theta_E|\boldsymbol{\theta}|$ in the text.

You'll need from before: nothing (calculus only).

(a) Compute ψ_{11}, ψ_{22} and ψ_{12} for the SIS. Write $\theta_1 = \theta\cos\phi$, $\theta_2 = \theta\sin\phi$.

(b) Show that $\kappa = \theta_E/(2\theta)$ and that $\gamma \equiv \gamma_1 + i\gamma_2 = -\frac{\theta_E}{2\theta}\,e^{2i\phi}$.

(c) Why does knowing κ everywhere fix γ everywhere (up to what)?

Solution. (a) $\partial_1\psi = \theta_E\theta_1/\theta$. Then $\psi_{11} = \theta_E\theta_2^2/\theta^3$, $\psi_{22} = \theta_E\theta_1^2/\theta^3$ and $\psi_{12} = -\theta_E\theta_1\theta_2/\theta^3$. (b) $\kappa = \frac{1}{2}\theta_E(\theta_1^2+\theta_2^2)/\theta^3 = \theta_E/(2\theta)$. $\gamma_1 = \theta_E(\theta_2^2-\theta_1^2)/(2\theta^3) = -\frac{\theta_E}{2\theta}\cos 2\phi$ and $\gamma_2 = -\frac{\theta_E}{2\theta}\sin 2\phi$, so $\gamma = -\frac{\theta_E}{2\theta}e^{2i\phi}$. (c) κ fixes $\nabla^2\psi$, which fixes ψ up to a harmonic function (one with $\nabla^2 = 0$, such as a constant gradient or a term $\propto\theta^2$ that is really a uniform sheet; see C3). The shear follows from ψ. This link is the basis of every mass map (Part D).

C2 Tangential and cross shear

Given. About the lens centre, the tangential and cross components of the shear are $\gamma_t = -\mathrm{Re}(\gamma\,e^{-2i\phi})$ and $\gamma_\times = -\mathrm{Im}(\gamma\,e^{-2i\phi})$.

Source: B&S01, Sec. 5.3.1 (Aperture Mass Measures), Eq. (5.25) for γ_t; Kilbinger15, Sec. 3.8 (The shear correlation function), Eq. (31) for γ_t and $\gamma_\times$. For any mass distribution, the tangential shear averaged around a circle of radius θ is $\langle\gamma_t\rangle(\theta) = \bar\kappa(\le\theta) - \langle\kappa\rangle(\theta)$, where $\bar\kappa(\le\theta)$ is the mean convergence inside the circle and $\langle\kappa\rangle(\theta)$ the mean on it. For an axially symmetric lens the averages can be dropped.

Source: Kilbinger15, Sec. 3.12 (Galaxy-galaxy lensing), Eq. (44); for $|\gamma|$ of an axisymmetric lens, B&S01, Sec. 3.1.5, footnote 5.

You'll need from before: C1 (SIS κ and γ).

(a) Use C1(b) to show that the SIS has $\gamma_t = \theta_E/(2\theta)$ and $\gamma_\times = 0$.

(b) Check the result against $\bar\kappa(<\theta) - \kappa(\theta)$.

(c) A subhalo next to a main halo breaks the axial symmetry about the main-halo centre. What does it add to $\gamma_\times$ measured about the main centre, and why do our polar cells stack in $(\gamma_t, \gamma_\times)$?

Solution. (a) $\gamma e^{-2i\phi} = -\frac{\theta_E}{2\theta}$ is real and negative, so $\gamma_t = \theta_E/(2\theta)$ and $\gamma_\times = 0$. (b) $\bar{\kappa}(<\theta) = \frac{1}{\pi\theta^2}\int_0^\theta \frac{\theta_E}{2\theta'}\, 2\pi\theta'\, d\theta' = \theta_E/\theta$, so $\bar{\kappa} - \kappa = \theta_E/\theta - \theta_E/(2\theta) = \theta_E/(2\theta)$. It agrees. (c) The subhalo's shear is tangential about *its own* centre, so about the main centre it has both γ_t and $\gamma_\times$ parts. A non-zero $\gamma_\times$ about the main centre signals non-axisymmetric structure. In the $(\gamma_t, \gamma_\times)$ frame the main halo's signal sits in one component, so departures from it stand out.

C3 Reduced shear and the mass-sheet degeneracy

Given. Galaxy shapes respond to the reduced shear $g = \gamma/(1-\kappa)$, not to γ.

Source: B&S01, Sec. 4.2 (Galaxy Shapes and Sizes, and their Transformation), Eq. (4.7); Kilbinger15, Sec. 3.3, Eq. (20). The transformation $\kappa \to \lambda\kappa + (1-\lambda)$, $\gamma \to \lambda\gamma$ (adding a uniform sheet of mass and rescaling) leaves g unchanged: the mass-sheet degeneracy.

Source: B&S01, Sec. 5.2.2 (Non-Linear Generalisation of the Inversion, and an Invariance Transformation), Eq. (5.9) and the text after it.

You'll need from before: C1.

(a) Verify that g is invariant under the transformation.

(b) For $\kappa = 0.05$ and $\gamma = 0.02$, compute g. How large is the difference from γ, in percent?

(c) What can shear data *not* tell you about a κ map?

Solution. (a) $g' = \lambda\gamma/(1 - \lambda\kappa - 1 + \lambda) = \lambda\gamma/[\lambda(1-\kappa)] = g$. (b) $g = 0.02/0.95 = 0.02105$, 5% above γ. That is negligible far out, but not near the cluster centre, where $\kappa \sim 0.2$. (c) The overall level (and scale) of κ: only its shape is fixed. A mass map's zero point is a convention.

C4 Adding two halos

Given. κ is the projected mass, and ψ obeys the linear Poisson equation $\nabla^2\psi = 2\kappa$. So the ψ, κ and γ of two halos are the sums of their separate values.

Source: B&S01, Sec. 3.1.2 (The Lens Equation), the Poisson equation stated just after Eq. (3.9); S96, Sec. 2.1 (Notation), Eqs. (3a)–(3b).

You'll need from before: C1, C3.

(a) Halo A gives $\kappa_A = 0.05$, $\gamma_A = (0.02, 0)$ at some point; halo B gives $\kappa_B = 0.03$, $\gamma_B = (0, 0.01)$. Compute the correct reduced shear of A+B.

(b) Compute the incorrect $g_A + g_B$. How big is the error?

(c) Why do our simulations add γ and κ over the halos first and form g only at the end?

Solution. (a) $\gamma = (0.02, 0.01)$ and $\kappa = 0.08$: $g = (0.02174, 0.01087)$. (b)

$$g_A + g_B = (0.02/0.95,\ 0.01/0.97) = (0.02105,\ 0.01031).$$

The error is $\approx$ 3–5%, as large as the effects we test for. (c) Because γ and κ superpose and g does not.

C5 Where the subhalo's signal is, and how strong it is

Given. In our simulation (subhalo of $3 \times 10^{14}\,M_\odot$ at $z = 0.13$), the subhalo's extra reduced shear at distance r from it is close to $|\Delta g| \approx A/r$ with $A \approx 0.009\,\mathrm{Mpc}$. That is SIS-like; the true profile is NFW, whose shear formulae are in WB00, Sec. 2 (Convergence and Shear of an NFW Object), Eqs. (11)–(16). The sources have density $n \approx 5\,\mathrm{Mpc}^{-2}$, and each measures each shear component with noise $\sigma \approx 0.05$. For a signal of known shape, the best possible (matched-filter) significance is $(\mathrm{S/N})^2 = \sum_i |\Delta\mathbf{g}_i|^2/\sigma^2$. *Derived here* (numbers from our `s3_sub1` manifest).

You'll need from before: A2 (information adds).

(a) Replace the sum by an integral over a uniform source density and show that $(\mathrm{S/N})^2 = 2\pi n A^2 \sigma^{-2} \ln(r_{\max}/r_{\min})$.

(b) Evaluate it for $r_{\min} = 0.1\,\mathrm{Mpc}$ and $r_{\max} = 3\,\mathrm{Mpc}$.

(c) What fraction of $(\mathrm{S/N})^2$ comes from within 1 Mpc, and from within 0.5 Mpc, of the subhalo?

(d) With each galaxy's full, anisotropic precision matrix in place of an isotropic σ, the same calculation gives $\mathrm{S/N} \approx 3.2$ for this galaxy set. Why is it larger?

Solution. (a) $\sum_i \to \int n\,2\pi r\,dr$: $(\mathrm{S/N})^2 = \int_{r_{\min}}^{r_{\max}} n\,2\pi r\,\frac{A^2}{r^2\sigma^2}\,dr = 2\pi n A^2\sigma^{-2}\ln(r_{\max}/r_{\min})$. (b) $A^2/\sigma^2 = 0.0324$, so $2\pi \cdot 5 \cdot 0.0324 = 1.02$, times $\ln 30 = 3.40$: $(\mathrm{S/N})^2 = 3.46$, so $\mathrm{S/N} \approx 1.9$. (The actual galaxies give 1.65 for isotropic $\sigma = 0.049$.) (c) Each factor of e in radius contributes equally: within 1 Mpc, $\ln 10/\ln 30 = 0.68$; within 0.5 Mpc, $\ln 5/\ln 30 = 0.47$. (The simulation gives 0.58 and 0.35; the NFW profile falls faster than $1/r$ at large r and is shallower near the centre.) The signal is spread over a few Mpc, not concentrated in one spot, which is why a method that uses every galaxy beats one cell. (d) KL measures one shear direction much better than the other (median $\sigma \approx 0.039$ along the best direction vs 0.057 along the worst). With matrix weights, each galaxy contributes most where it is precise (A3). This is the benchmark for Part G: no map can beat the matched filter.

Part D. Fourier space and Kaiser–Squires

D1 Derivatives become multiplications

Given. With the Fourier transform $\hat{f}(\mathbf{k}) = \int f(\boldsymbol{\theta})\,e^{i\mathbf{k}\cdot\boldsymbol{\theta}}\,d^2\theta$ (the sign convention of B&S01, Sec. 5.2.1, Eq. (5.1)), a derivative $\partial/\partial\theta_j$ of f becomes a factor $\mp ik_j$ on $\hat{f}$. The sign depends on the convention and will cancel in everything below, because every quantity uses *two* derivatives. The potential obeys the Poisson equation $\nabla^2\psi = 2\kappa$.

Source: B&S01, Sec. 3.1.2 (The Lens Equation), the Poisson equation stated just after Eq. (3.9); S96, Sec. 2.1 (Notation), Eqs. (3a)–(3b).

You'll need from before: C1 (κ and γ as second derivatives of ψ).

(a) Show that $\widehat{\partial_i\partial_j f} = -k_i k_j \hat{f}$ for either sign convention.

(b) Show that $\hat{\psi} = -2\hat{\kappa}/k^2$ for $\mathbf{k} \neq 0$.

(c) What happens at $\mathbf{k} = 0$? Relate this to C3.

Solution. (a) Each derivative gives $\mp ik$; two give $(\mp i)^2 k_i k_j = -k_i k_j$. (b) $\nabla^2 \to -(k_1^2 + k_2^2) = -k^2$, so $-k^2\hat{\psi} = 2\hat{\kappa}$. (c) The equation says nothing about $\hat{\kappa}(0)$, the mean κ over the field: shear never constrains it. It is the mass-sheet degeneracy of C3 in Fourier form.

D2 The Kaiser–Squires kernel

Given. The shear is a convolution of κ with a kernel. In Fourier space, one multiplication per mode links them, and the inversion recovers $\hat{\kappa}$ from $\hat{\gamma}$ by the complex-conjugate kernel. The papers write the kernel as $\hat{D} = \pi(k_1^2 - k_2^2 + 2ik_1k_2)/k^2$ with a factor π^{-1} in front of the product; below we absorb the π.

Source: B&S01, Sec. 3.1.3, Eq. (3.15) (forward) and Sec. 5.2.1 (Linear Inversion of Shear Maps), Eqs. (5.1)–(5.4) (inverse); KS93, Sec. 2.1 (Idealized Analysis), Eqs. (2.1.12) and (2.1.15); J18, Sec. 2.2.1 (Theory), Eqs. (12)–(14).

You'll need from before: C1, D1.

(a) Using C1 and D1, show that $\hat{\gamma} = D\,\hat{\kappa}$ with $\hat{\gamma} = \hat{\gamma}_1 + i\hat{\gamma}_2$ and $D(\mathbf{k}) = (k_1^2 - k_2^2 + 2ik_1k_2)/k^2$.

(b) Show that $|D| = 1$. Write $D = e^{2i\beta}$, with β the direction of $\mathbf{k}$.

(c) Show that $\hat{\kappa} = D^*\hat{\gamma}$ for $\mathbf{k} \neq 0$. Why is this inversion exact for noise-free data on an infinite field, yet noisy in practice?

Solution. (a) $\hat{\gamma}_1 = \frac{1}{2}(\widehat{\psi_{11}} - \widehat{\psi_{22}}) = \frac{1}{2}(-k_1^2 + k_2^2)\hat{\psi} = \frac{k_1^2 - k_2^2}{k^2}\hat{\kappa}$ and $\hat{\gamma}_2 = \widehat{\psi_{12}} = -k_1k_2\hat{\psi} = \frac{2k_1k_2}{k^2}\hat{\kappa}$. (b) $|D|^2 = [(k_1^2 - k_2^2)^2 + 4k_1^2k_2^2]/k^4 = (k_1^2 + k_2^2)^2/k^4 = 1$. With $\mathbf{k} = k(\cos\beta, \sin\beta)$, $D = \cos 2\beta + i \sin 2\beta = e^{2i\beta}$. (c) $D^*D = |D|^2 = 1$. Noise $\hat{n}$ in the shear map passes into $\hat{\kappa}$ undamped ($|D^*| = 1$) at every scale, including small scales where the signal is tiny. KS therefore needs smoothing afterwards (D4).

D3 Finite fields, padding and the mean mass

Given. The inversion holds for $\mathbf{k} \neq 0$; the real-space form carries an undetermined additive constant κ_0, because a uniform surface mass density produces no shear.

Source: B&S01, Sec. 3.1.3, Eq. (3.15) (forward) and Sec. 5.2.1 (Linear Inversion of Shear Maps), Eqs. (5.1)–(5.4) (inverse); KS93, Sec. 2.1 (Idealized Analysis), Eqs. (2.1.12) and (2.1.15); B&S01, Sec. 5.2.2 (Non-Linear Generalisation of the Inversion, and an Invariance Transformation), Eq. (5.9) and the text after it. On a grid, the discrete Fourier transform treats the field as periodic.

You'll need from before: D1, D2.

(a) Why does a periodic transform make the left edge of a shear map act on the right edge? What does padding the map with zeros to twice its size do?

(b) Our KS map sets $\hat{\kappa}(0) = 0$. What does that assume, and can the data test it?

(c) Our likelihood κ map pads with zeros too, in the forward direction ($\kappa \to \gamma$). What assumption does that make about κ outside the field?

Solution. (a) In a periodic box, mass near one edge is also "next to" the opposite edge, and the non-local kernel (shear falls only as $1/r^2$) couples them. Padding with zeros places the copies far away and removes most of that wrap-around. (b) That the mean κ in the field is zero. The data cannot test it (D1(c)), so the map's zero point is a convention. (c) That $\kappa = 0$ outside the field. Mass outside still shears galaxies inside, so this is an approximation that matters most near the edge.

D4 [Py] What binning does to a tangential pattern

Given. A lens at the origin produces purely tangential shear, $g = -g_t(r)\, e^{2i\phi}$ (B5(b)). KS needs the shear on a regular grid, so it averages the galaxies in each cell. Smoothing by a Gaussian of width s multiplies each Fourier mode by $e^{-k^2 s^2/2}$ (the convolution theorem). *Derived here*; the cell-averaging argument is ours. Our cells are 0.98 Mpc squares and our subhalo sits almost at a cell centre.

You'll need from before: B5(b), D2.

(a) Show that the average of g_1 and g_2 over a square centred on the lens is zero for any $g_t(r)$. (Use the symmetry of $\cos 2\phi$ and $\sin 2\phi$ under $\phi \to \phi + \pi/2$ and $\phi \to -\phi$.)

(b) So what does the KS cell containing the subhalo report about the subhalo? Where does KS's information about the subhalo come from?

(c) [Py] Compute the cell-averaged shear of an SIS-like subhalo ($g_t = A/r$) for cells whose centre is offset from the subhalo by 0 to 1 cell, and compare with the subhalo's shear at the cell centre.

(d) Gaussian smoothing with $s = 0.7$ cell multiplies the mode at the grid's Nyquist scale ($k = \pi/\Delta$, cell size Δ) by what factor? What does that mean for the map's resolution?

Solution. (a) The square is symmetric under a 90° rotation, which sends $\cos 2\phi \to -\cos 2\phi$ and $\sin 2\phi \to -\sin 2\phi$ while r is unchanged, so each average equals minus itself. (b) Nothing (to first order): the cell centred on the subhalo averages its signal away. KS learns about the subhalo only from the neighbouring cells, where the pattern no longer cancels. The likelihood map, which compares each galaxy with the shear at its own position, keeps that signal. (c) See the notebook: the cell average is zero at offset 0 and grows to about the centre value once the cell is about one cell width away. The nearest cells are where the cancellation hurts. (d) $e^{-(\pi\cdot 0.7)^2/2} = e^{-2.42} = 0.089$. Structure at the scale of one or two cells is suppressed about tenfold, so KS resolves only features several cells (several Mpc) across.

Part E. Regularised inversion and the Wiener filter

E1 When the data can't determine the answer

Given. A linear model $\mathbf{d} = A\mathbf{s} + \mathbf{n}$, with unknowns $\mathbf{s}$ and noise covariance N. Weighted least squares gives $\hat{\mathbf{s}} = [A^\mathsf{T} N^{-1} A]^{-1} A^\mathsf{T} N^{-1} \mathbf{d}$, which needs $A^\mathsf{T} N^{-1} A$ to be invertible.

Source: HBL10, Sec. 1 (Standard practice), Eqs. (5) and (7), with the parameter covariance $[A^\mathsf{T} C^{-1} A]^{-1}$ stated in the text after Eq. (7).

You'll need from before: A3 (the matrix least-squares solution).

(a) One datum measures the sum of two unknowns: $d = s_1 + s_2 + n$, so $A = (1 \;\; 1)$. Show that $A^\mathsf{T} A$ is singular. Describe all the solutions that fit the datum perfectly.

(b) Two data, $A = \mathrm{diag}(1, 0.01)$, $N = \sigma^2 I$. Show that $\hat{s}_2$ amplifies the noise 100-fold.

(c) Our κ map has 144 unknown pixels and about 150 galaxies, each with noise ≈ 0.05 against a subhalo signal of ≈ 0.01. Which of (a) and (b) describes it?

Solution. (a) $A^\mathsf{T} A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has determinant 0. Any (s_1, s_2) with $s_1 + s_2 = d$ fits; the data fix one combination and say nothing about $s_1 - s_2$. (b) $\hat{s}_2 = d_2/0.01 = s_2 + 100\, n_2$: a direction the data barely respond to is recovered with 100 times the noise. (c) Both. Small-scale patterns of κ produce shears that nearly cancel at the galaxies (small singular values of A, as in (b)), and the noisy, nearly degenerate system has many near-equally good solutions (as in (a)). Something other than the data must choose between them: a prior.

E2 A Gaussian prior makes the problem well posed

Given. Add a Gaussian prior $\mathbf{s} \sim \mathcal{N}(0, S)$ to the model of E1. For a Gaussian signal and Gaussian noise, the linear minimum-variance (Wiener) filter is $W = SA^\mathsf{T}(ASA^\mathsf{T} + N)^{-1}$, and it is also the maximum and the mean of the Gaussian posterior.

Source: Z95, Sec. 2b (Wiener Filter), Eqs. (2–11a,b), and Sec. 2c (Conditional Probability), the statement $s_{\mathrm{MV}} = s_{\mathrm{mean}} = s_{\mathrm{MAP}}$; J18, Sec. 2.3.1 (Theory), Eqs. (17)–(18) (linear minimum-variance filter) and Eqs. (19)–(21) (the MAP solution for a Gaussian prior and Gaussian noise).

You'll need from before: A3 (completing the square in matrix form), E1.

(a) Write $-2 \ln p(\mathbf{s} \mid \mathbf{d}) = (\mathbf{d} - A\mathbf{s})^\mathsf{T} N^{-1} (\mathbf{d} - A\mathbf{s}) + \mathbf{s}^\mathsf{T} S^{-1} \mathbf{s} + \mathrm{const}$. Complete the square to show that the posterior is Gaussian with precision $\mathsf{H} = S^{-1} + A^\mathsf{T} N^{-1} A$ and mean $\hat{\mathbf{s}} = \mathsf{H}^{-1} A^\mathsf{T} N^{-1} \mathbf{d}$.

(b) Show that $\hat{\mathbf{s}} = W\mathbf{d}$, using the identity $(S^{-1} + A^\mathsf{T} N^{-1} A)^{-1} A^\mathsf{T} N^{-1} = SA^\mathsf{T}(ASA^\mathsf{T} + N)^{-1}$. Verify the identity by multiplying both sides on the left by $S^{-1} + A^\mathsf{T} N^{-1} A$ and on the right by $ASA^\mathsf{T} + N$.

(c) Redo E1(a) with $S = \sigma_s^2 I$ and $N = \sigma_n^2$. Is H invertible now? What is $\hat{\mathbf{s}}$?

Solution. (a) The $\mathbf{s}$-dependent terms are $\mathbf{s}^\mathsf{T}(A^\mathsf{T}N^{-1}A + S^{-1})\mathbf{s} - 2\mathbf{s}^\mathsf{T}A^\mathsf{T}N^{-1}\mathbf{d}$, which equals $(\mathbf{s}-\hat{\mathbf{s}})^\mathsf{T}\mathsf{H}(\mathbf{s}-\hat{\mathbf{s}}) + \text{const}$ with $\hat{\mathbf{s}} = \mathsf{H}^{-1}A^\mathsf{T}N^{-1}\mathbf{d}$. A Gaussian's mean and maximum coincide. (b) Left side times the two factors: $A^\mathsf{T}N^{-1}(ASA^\mathsf{T} + N) = A^\mathsf{T}N^{-1}ASA^\mathsf{T} + A^\mathsf{T}$. Right side: $(S^{-1} + A^\mathsf{T}N^{-1}A)SA^\mathsf{T} = A^\mathsf{T} + A^\mathsf{T}N^{-1}ASA^\mathsf{T}$. They agree. (c) $\mathsf{H} = \sigma_s^{-2}I + \sigma_n^{-2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$ has eigenvalues $\sigma_s^{-2} + 2\sigma_n^{-2}$ and σ_s^{-2}, both positive, so it is invertible. By symmetry $\hat{s}_1 = \hat{s}_2 = \frac{d\,\sigma_s^2}{2\sigma_s^2+\sigma_n^2}$: the datum is split equally, shrunk toward zero, and the undetermined direction $s_1 - s_2$ is set to its prior value, 0.

E3 One mode at a time: what the Wiener filter keeps

Given. If S and N are diagonal in the same basis, for example Fourier modes of a stationary field, the filter acts on each mode separately and suppresses the noise by the ratio prior/(prior + noise). It is the linear estimate with the smallest residual variance.

Source: Z95, Sec. 2b, Eqs. (2–3)–(2–5) (minimum residual variance); the single-mode ratio prior/(prior + noise) is in the Sec. 2b text, and per harmonic in Sec. 3b, Eq. (3–9).

You'll need from before: E2.

(a) One mode, $d = s + n$, with $s \sim \mathcal{N}(0, S)$ and $n \sim \mathcal{N}(0, N)$. For an estimate $\hat{s} = w\,d$, show that the mean squared error $\mathbb{E}[(\hat{s} - s)^2] = (1-w)^2S + w^2N$, and that it is minimised at $w = S/(S+N)$.

(b) For a *fixed* true s, show that $\mathbb{E}[\hat{s} \mid s] = ws$. In what sense is the Wiener estimate biased?

(c) Large scales have $S \gg N$ and small scales $S \ll N$. Describe the filtered map.

(d) The subhalo in our likelihood map recovers about 40% of its true peak value. What value of w does that suggest for the scales it occupies?

Solution. (a) $\hat{s} - s = (w-1)s + wn$; with s and n independent, the variance is $(1-w)^2S + w^2N$. Setting the derivative to zero, $-2(1-w)S + 2wN = 0$, gives $w = S/(S+N)$. (b) $\mathbb{E}[w(s+n) \mid s] = ws$: every mode is shrunk toward zero by $w < 1$. The estimate is biased for any given field, but over the ensemble of fields its error is minimal. This is the trade made on purpose. (c) Large scales pass almost unchanged and small scales are suppressed, so the map is a smoothed, slightly shrunk version of the truth. (d) $w \approx 0.4$: at the subhalo's scale the signal and noise powers are comparable ($S/N \approx 0.7$).

E4 A smoothness penalty is a power spectrum

Given. Our map prior is $-\ln p(\kappa) = \frac{1}{2\sigma_\kappa^2}\big[\sum_j \kappa_j^2 + \ell^2 \sum_{\langle jj'\rangle}(\kappa_j - \kappa_{j'})^2\big]$, summed over pixels j and neighbouring pairs $\langle jj'\rangle$. *Derived here*; it is the regulariser of `likelihood_kappa` in `h_cluster_analysis.py`.

You'll need from before: E2, E3 (a Gaussian prior with covariance S; per-mode filtering).

(a) On a periodic 1D chain of M pixels, insert a Fourier mode $\kappa_j = a\, e^{ikj}$. Show that $\sum_j |\kappa_{j+1} - \kappa_j|^2 = 4\sin^2(k/2)\sum_j |\kappa_j|^2$.

(b) Conclude that each mode has prior variance $P(k) = \sigma_\kappa^2/[1 + 4\ell^2 \sin^2(k/2)] \approx \sigma_\kappa^2/(1 + \ell^2 k^2)$ for small k.

(c) What do σ_κ and ℓ each control? Which scales does the prior suppress?

Solution. (a) $\kappa_{j+1} - \kappa_j = a\, e^{ikj}(e^{ik} - 1)$ and $|e^{ik} - 1|^2 = 2 - 2\cos k = 4\sin^2(k/2)$. (b) Modes are independent (they diagonalise the quadratic form), and mode k has precision $[1 + 4\ell^2 \sin^2(k/2)]/\sigma_\kappa^2$, whose inverse is the prior variance. For small k, $4\sin^2(k/2) \approx k^2$. (c) σ_κ sets the overall amplitude the prior allows. ℓ sets the scale below which fluctuations are suppressed: the variance falls by half at $k \approx 1/\ell$. Combined with E3, the Wiener weight per mode is $P(k)/[P(k) + N(k)]$, which removes small scales first.

E5 [Py] A one-dimensional Wiener filter

Given. A toy 1D "κ" made of a broad bump (the main halo) plus a narrow bump (the subhalo) is sampled by noisy point measurements at random positions, each with noise $\sigma = 0.05$ of a quantity linear in κ. The MAP map under the prior of E4 is $\hat{\kappa} = \mathsf{H}^{-1} A^\mathsf{T} N^{-1} \mathbf{d}$ (E2). *Derived here.*

You'll need from before: E2, E4.

(a) [Py] Build A (a measurement samples the pixel it falls in) and the prior precision of E4. Compute the MAP map for $\sigma_\kappa \in \{0.03, 0.1, 0.3\}$ and $\ell \in \{0.5, 1, 2\}$ pixels.

(b) [Py] Which settings recover the narrow bump best? What happens to it for small σ_κ, and for large ℓ?

(c) [Py] Repeat with 20 noise realizations. For one setting, plot the mean map against the truth. Is the mean map equal to the truth?

Solution. (a)–(b) See the solutions notebook. Small σ_κ shrinks everything (low peaks); large ℓ blurs the narrow bump into the broad one. Large σ_κ with small ℓ keeps the bump's height but lets the noise through. The best setting balances the two, just as the tuning did on the real map. (c) No: the mean map is a smoothed, shrunk version of the truth (E3(b)). Averaging over noise removes the noise, not the prior's bias. Part F makes this precise with the resolution matrix.

Part F. Uncertainty, resolution and effective parameters

F1 The Laplace approximation

Given. Near its maximum $\mathbf{x}_0$, a log density is approximately quadratic, $\ln P(\mathbf{x}) \simeq \ln P(\mathbf{x}_0) - \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\mathsf{T} \mathsf{A} (\mathbf{x} - \mathbf{x}_0)$, with $\mathsf{A} = -\nabla\nabla \ln P|_{\mathbf{x}_0}$. Treating P as a Gaussian with covariance A^{-1} is the Laplace approximation.

Source: ITILA03, Ch. 27 (Laplace's Method), Eqs. (27.6)–(27.8).

You'll need from before: A1 (a Gaussian's log is a quadratic).

(a) For $P(x) \propto x^a e^{-bx}$ $(x > 0)$, find the mode, A, and the Laplace standard deviation.

(b) The exact standard deviation is $\sqrt{a+1}/b$. Compare for $a = 4$. When does Laplace work well?

(c) For the posterior of E2, show that the Laplace approximation is exact, with $\mathsf{A} = \mathsf{H}$.

Solution. (a) $\ln P = a \ln x - bx$; the mode is $x_0 = a/b$, $\mathsf{A} = a/x_0^2 = b^2/a$, so $\sigma_L = \sqrt{a}/b$. (b) $a = 4$: $2/b$ against $\sqrt{5}/b = 2.24/b$, 11% low. It works well when the distribution is close to Gaussian, here for large a (the ratio $\sqrt{a/(a+1)} \to 1$). (c) The log posterior is exactly quadratic, with $-\nabla\nabla \ln p = S^{-1} + A^{\mathsf{T}} N^{-1} A = \mathsf{H}$.

F2 Gauss–Newton: curvature from the model, not from the data grid

Given. For a least-squares objective $F(\mathbf{x}) = \frac{1}{2}\sum_i f_i(\mathbf{x})^2$, the exact Hessian is $J^{\mathsf{T}}J + \sum_i f_i \nabla\nabla f_i$, with J the Jacobian of the residuals f_i. Gauss–Newton keeps only $J^{\mathsf{T}}J$.

Source: MNT04, Ch. 3, Eq. (3.4b) (the exact Hessian) and Sec. 3.1 (The Gauss–Newton Method), Eq. (3.8). They treat unweighted least squares; the weighted form $J^{\mathsf{T}}FJ$ is derived here. Our galaxies enter through $\ln L_i(\mathbf{g}_i(\kappa))$, read off a grid by bilinear interpolation, and each L_i has precision F_i (A3).

You'll need from before: A3, F1.

(a) Suppose $\ln L_i(\mathbf{g}) \approx -\frac{1}{2}(\mathbf{g} - \hat{\mathbf{g}}_i)^{\mathsf{T}} F_i (\mathbf{g} - \hat{\mathbf{g}}_i)$. Show that the Hessian of $-\sum_i \ln L_i(\mathbf{g}_i(\kappa))$ with respect to the pixels is $\sum_i J_i^{\mathsf{T}} F_i J_i +$ (terms $\propto$ residuals $\times \nabla\nabla \mathbf{g}_i$), with $J_i = \partial \mathbf{g}_i / \partial \kappa$.

(b) Why are the dropped terms small for us? (Think about how $\mathbf{g}_i$ depends on κ; see Part D.)

(c) A bilinear interpolant is linear along each axis inside a grid cell. What is its second derivative along g_1 there? Why does that make the *exact* Hessian of our objective useless, and why does $J^{\mathsf{T}}FJ$ not have this problem?

Solution. (a) The gradient is $-\sum_i J_i^{\mathsf{T}} \nabla_{\mathbf{g}} \ln L_i = \sum_i J_i^{\mathsf{T}} F_i (\mathbf{g}_i - \hat{\mathbf{g}}_i)$. Differentiating again gives $\sum_i J_i^{\mathsf{T}} F_i J_i$ plus terms with $\partial J_i / \partial \kappa$ multiplied by the residual $F_i(\mathbf{g}_i - \hat{\mathbf{g}}_i)$. This is the weighted version of MNT04's Eq. (3.4b). (b) $\mathbf{g}_i$ is linear in κ except for the $1/(1-\kappa)$ factor (Part D), so $\nabla\nabla \mathbf{g}_i$ is small, and it is multiplied by residuals that scatter around zero. (c) Zero. Inside each grid cell the interpolant has no curvature along g_1 or g_2; the curvature lives only in the kinks between cells. The exact Hessian therefore misses most of the data's information and can even have negative directions (in our first attempt, 12 pixels got $\sigma = 0$). $J^{\mathsf{T}}FJ$ takes the curvature from each galaxy's known precision F_i, which is positive definite by construction.

F3 Two different uncertainties, and the resolution matrix

Given. In the linear-Gaussian model of E2, write $\mathsf{H}_d = A^{\mathsf{T}} N^{-1} A$ (data) and $\mathsf{H}_p = S^{-1}$ (prior), so $\mathsf{H} = \mathsf{H}_d + \mathsf{H}_p$ and $\hat{\mathbf{s}} = \mathsf{H}^{-1} A^{\mathsf{T}} N^{-1} \mathbf{d}$. Let the truth be $\mathbf{s}_{\text{true}}$, with $\mathbf{d} = A\mathbf{s}_{\text{true}} + \mathbf{n}$. *Derived here* (`KAPPA_FITTING_EXPLAINED.md`, Sec. 8.3).

You'll need from before: E2, F1.

(a) Show that $\mathbb{E}_{\mathbf{n}}[\hat{\mathbf{s}}] = \mathsf{R}\,\mathbf{s}_{\text{true}}$ with $\mathsf{R} = \mathsf{H}^{-1}\mathsf{H}_d$.

(b) Show that the covariance of $\hat{\mathbf{s}}$ over noise realizations is $\mathsf{H}^{-1}\mathsf{H}_d\mathsf{H}^{-1}$.

(c) Scalar case ($\mathsf{H}_d = h$, $\mathsf{H}_p = p$): compare this noise variance with the posterior variance $1/(h+p)$. Which is larger, and what does each one answer?

(d) On our map, the noise covariance predicts an aperture scatter of 0.0034 and the posterior covariance 0.0067. In the scalar picture, what data-to-prior ratio h/p does that imply? For "is this blob a noise fluctuation?", which one should the S/N map use?

Solution. (a) $\mathbb{E}[\mathbf{d}] = A\mathbf{s}_{\text{true}}$, so

$$\mathbb{E}[\hat{\mathbf{s}}] = \mathsf{H}^{-1}A^{\mathsf{T}}N^{-1}A\,\mathbf{s}_{\text{true}} = \mathsf{H}^{-1}\mathsf{H}_d\,\mathbf{s}_{\text{true}}.$$

(b) Cov $= \mathsf{H}^{-1}A^{\mathsf{T}}N^{-1}\,N\,N^{-1}A\mathsf{H}^{-1} = \mathsf{H}^{-1}\mathsf{H}_d\mathsf{H}^{-1}$. (c) Noise variance $h/(h+p)^2$ = posterior variance $\times\, h/(h+p)$, so it is always smaller. The posterior variance answers "how unsure am I about the true value?", including what the prior allows. The noise variance answers "how much would this map change with new noise?". (d) $(0.0034/0.0067)^2 = 0.26 = h/(h+p)$, so $h/p \approx 0.35$: on the subhalo's scale the prior outweighs the data about 3 to 1. The S/N map should use the noise covariance, which is what checks out against the scatter over synthetic realizations (0.0029).

F4 [Py] How many numbers do the data actually determine?

Given. In the eigenbasis of the data Hessian (eigenvalues λ_a), with an isotropic prior of precision α, the number of parameters well determined by the data is $\gamma = \sum_a \lambda_a/(\lambda_a + \alpha)$. Each term, between 0 and 1, measures the data's strength relative to the prior in that direction.

Source: M92t, Ch. 2 (Bayesian Interpolation), Sec. 2.4, Eq. (2.23); the same result is M92, Sec. 4, Eq. (23) of the preprint.

You'll need from before: F3 (the resolution matrix R), E5 (the 1D toy problem).

(a) Show that $\operatorname{tr}\mathsf{R} = \gamma$ when $\mathsf{H}_p = \alpha I$. (Diagonalise H_d.)

(b) [Py] For the 1D problem of E5, compute $\operatorname{tr}\mathsf{R}$ as a function of the number of pixels (keeping the physical ℓ fixed). Does it keep growing?

(c) Our tuned map has 144 pixels and about 150 galaxies, and $\operatorname{tr}\mathsf{R} = 16.5$. With a weaker prior ($\sigma_\kappa = 0.1$) it is 53; with 576 pixels it is 10. Answer the question "how can 141 galaxies fit 144 pixels?"

(d) What sets the map's effective resolution, if not the pixel size?

Solution. (a) With $\mathsf{H}_d = U\Lambda U^{\mathsf{T}}$, $\mathsf{R} = (\mathsf{H}_d + \alpha I)^{-1}\mathsf{H}_d = U\,\operatorname{diag}\big(\lambda_a/(\lambda_a+\alpha)\big)U^{\mathsf{T}}$, whose trace is γ. (b) No. In the notebook $\operatorname{tr}\mathsf{R}$ rises to ≈ 19 near 60 pixels and then *falls* (7.7 at 240). Extra pixels add only directions the data cannot see. And with σ_κ fixed per pixel, the amplitude term $\sum \kappa_j^2$ runs over more pixels, so every smooth pattern is penalised more: a finer grid with the same σ_κ is a stronger prior. (c) They don't, and they don't need to. It is a regularised fit: the galaxies pin down about 16 independent patterns, and the prior fixes the remaining ~ 128. A weaker prior lets the data set more patterns, but noisily. More pixels add only prior-dominated directions (and, at

fixed σ_κ, a stronger prior), as the drop to 10 at 576 pixels shows. (d) The rows of R: their width is the resolution and their height the shrinkage, both set jointly by the galaxies (number, positions, precisions) and by the prior. The pixel size only sets the finest scale the map could show.

Part G. The cluster analysis

G1 Bernstein & Armstrong for a constant shear

Given. BA14 write the posterior of one shear shared by many galaxies as $P(\mathbf{g} \mid D) \propto P(\mathbf{g}) \prod_i P(D_i \mid \mathbf{g})$. They expand each galaxy's likelihood to second order about $\mathbf{g} = 0$, $P(D_i \mid \mathbf{g}) \approx P_i + \mathbf{g} \cdot \mathbf{Q}_i + \frac{1}{2}\mathbf{g} \cdot \mathbf{R}_i \cdot \mathbf{g}$ (here $\mathbf{R}_i$ is BA14's second-derivative matrix, not our rotation), and find a Gaussian posterior with $C_g^{-1} = \sum_i (\mathbf{Q}_i \mathbf{Q}_i^\mathsf{T}/P_i^2 - \mathbf{R}_i/P_i)$ and $\bar{\mathbf{g}} = C_g \sum_i \mathbf{Q}_i/P_i$.

Source: BA14, Sec. 2.1 (General formulation), Eqs. (2)–(7).

You'll need from before: A3, A4, B3, B4.

(a) Show that $\mathbf{Q}_i/P_i = \nabla \ln P(D_i \mid \mathbf{g})|_0$ and $\mathbf{Q}_i\mathbf{Q}_i^\mathsf{T}/P_i^2 - \mathbf{R}_i/P_i = -\nabla\nabla \ln P(D_i \mid \mathbf{g})|_0 \equiv \mathbf{A}_i$.

(b) For a Gaussian likelihood $\ln L_i = -\frac{1}{2}(\mathbf{g} - \hat{\mathbf{g}}_i)^\mathsf{T} F_i (\mathbf{g} - \hat{\mathbf{g}}_i)$, show that $\mathbf{A}_i = F_i$ and $\nabla \ln L_i|_0 = F_i \hat{\mathbf{g}}_i$, and hence that BA14's $\bar{\mathbf{g}}$ is our stack $F_{\text{tot}}^{-1} \sum_i F_i \hat{\mathbf{g}}_i$ (A3).

(c) Our likelihood is $L_i = q_i/\pi_{\text{train}}$ (A4) on a 401^2 grid, multiplied without Taylor expansion. What do we gain over the quadratic form, and what does the Gaussian form still do for us?

Solution. (a) $\nabla \ln P = \nabla P/P$, which at 0 is $\mathbf{Q}_i/P_i$. $\nabla\nabla \ln P = \nabla\nabla P/P - \nabla P \nabla P^\mathsf{T}/P^2$, which at 0 is $\mathbf{R}_i/P_i - \mathbf{Q}_i\mathbf{Q}_i^\mathsf{T}/P_i^2$. (b) $\nabla \ln L_i = -F_i(\mathbf{g} - \hat{\mathbf{g}}_i)$, which at 0 is $F_i\hat{\mathbf{g}}_i$, and $-\nabla\nabla \ln L_i = F_i$. So $C_g^{-1} = \sum_i F_i$ and $\bar{\mathbf{g}} = (\sum_i F_i)^{-1} \sum_i F_i \hat{\mathbf{g}}_i$. (c) The full product is exact at any $|g| < 0.1$ and includes the higher-order terms BA14 add as perturbations (BA14, Appendix A (Third-order posterior calculation)). The Gaussian form still defines the reference shear: BA14's $\mathbf{A}_i$ is our F_i, so $\mathbf{g}_{\text{ref}}$ uses the same weights the stack does (B4).

G2 Bernstein & Armstrong for a varying shear

Given. For any model with parameters $\mathbf{a}$ that predicts a shear $\mathbf{g}_i(\mathbf{a})$ at each galaxy, BA14's quadratic expansion gives (written per galaxy)

$$-\ln P(\mathbf{a} \mid D) \approx \text{const} - \ln P(\mathbf{a}) - \sum_i \mathbf{g}_i(\mathbf{a}) \cdot \frac{\mathbf{Q}_i}{P_i} + \frac{1}{2} \sum_i \mathbf{g}_i(\mathbf{a}) \cdot \mathbf{A}_i \cdot \mathbf{g}_i(\mathbf{a}).$$

If $\mathbf{g}_i$ is linear in $\mathbf{a}$ and the prior is Gaussian, the maximum has a closed form.

Source: BA14, Sec. 2.2 (Non-constant shear), Eqs. (8)–(10).

You'll need from before: G1, B4, E2, E4.

(a) *Cells.* Let $\mathbf{a}$ be one shear per cell, with $\mathbf{g}_i(\mathbf{a}) = R_i^\mathsf{T} \mathbf{G}_{c(i)}$. Show that the posterior separates into independent cell stacks. Using $\mathbb{E}[\mathbf{Q}_i/P_i] \approx \mathbf{A}_i \mathbf{t}_i$ (the linear response of B4), show that the stack converges to $\mathbf{g}_{\text{ref}}$.

(b) *κ map.* Let $\mathbf{a}$ be the κ pixels, with $\mathbf{g}_i(\kappa)$ from Part D. Neglecting the $1/(1-\kappa)$ factor, $\mathbf{g}_i$ is linear in κ: write $\mathbf{g} = M\kappa$. With the prior of E4, which closed form does BA14's statement give, and what is it called?

(c) *Halo fit.* What is $\mathbf{a}$, and is the model linear?

(d) In one sentence: what does a galaxy contribute to any shear-field model?

Solution. (a) Each galaxy's term depends only on its own cell's $\mathbf{G}_c$, so the sum splits by cell, giving one A3/G1 stack per cell. Setting the gradient with respect to $\mathbf{G}_c$ to zero, $\sum_{i\in c}\left(R_i\mathbf{A}_iR_i^{\mathsf{T}}\mathbf{G}_c - R_i\mathbf{Q}_i/P_i\right) = 0$, so $\hat{\mathbf{G}}_c = (\sum_i F_i')^{-1}\sum_i R_i\mathbf{Q}_i/P_i$ with $F_i' = R_i\mathbf{A}_iR_i^{\mathsf{T}}$. With $\mathbb{E}[\mathbf{Q}_i/P_i] = \mathbf{A}_i\mathbf{g}_i = \mathbf{A}_iR_i^{\mathsf{T}}\mathbf{t}_i$, each term becomes $F_i'\mathbf{t}_i$, and $\mathbb{E}[\hat{\mathbf{G}}_c] = (\sum_i F_i')^{-1}\sum_i F_i'\mathbf{t}_i = \mathbf{g}_{\rm ref}$. (b) Write the linear forward model as $\mathbf{g} = M\kappa$ (M stacks the rows $\partial\mathbf{g}_i/\partial\kappa$), let $\mathcal{A} = \mathrm{blockdiag}(\mathbf{A}_i)$ and $\mathbf{q}$ stack the $\mathbf{Q}_i/P_i$, and take the Gaussian prior S^{-1} of E4. The objective is quadratic, so $\hat{\kappa} = (S^{-1} + M^{\mathsf{T}}\mathcal{A}M)^{-1}M^{\mathsf{T}}\mathbf{q}$. This is E2's estimator with $\mathcal{A}$ in the role of N^{-1} (and $\mathbf{q} = \mathcal{A}\hat{\mathbf{g}}$ for Gaussian likelihoods): the Wiener filter. Our map is its non-linear, non-Gaussian generalisation, solved numerically. (c) $\mathbf{a} = (M_{\rm main}, M_{\rm sub})$; the NFW shear is non-linear in the masses, so it needs sampling or optimisation. (d) Its log-likelihood evaluated at the shear the model predicts at that galaxy's position: no per-galaxy "shear estimate" is ever needed.

G3 Why the difference map is clean

Given. In each realization, the main-only and main + sub datasets have the same galaxies and the same noise draws; only the true shears differ. The map estimator is (nearly) the linear MAP of E2, $\hat{\kappa} = \mathsf{H}^{-1}A^{\mathsf{T}}N^{-1}\mathbf{d}$. The difference map is $\hat{\kappa}_{\rm main+sub} - \hat{\kappa}_{\rm main}$. *Derived here* (`KAPPA_FITTING_EXPLAINED.md`, Sec. 9.4).

You'll need from before: E2, F3, F4.

(a) In the linear model, show that the difference map equals $\mathsf{R}\,\kappa_{\rm subhalo}$ *exactly*, with no noise at all.

(b) Our real difference maps have a background rms of ≈ 0.002 against a peak of ≈ 0.014. Where can noise come from, if (a) says there is none?

(c) Why can't a real observation make a difference map, and what does it show instead?

(d) The subhalo's true peak pixel is 0.030 and the difference map peaks at about 0.012–0.014. Explain with R.

Solution. (a) $\mathbf{d}_{\rm sub} - \mathbf{d}_{\rm main} = A(\kappa_{\rm sub} - \kappa_{\rm main}) = A\kappa_{\rm subhalo}$, because the noise is identical and cancels. Then the difference map is $\mathsf{H}^{-1}A^{\mathsf{T}}N^{-1}A\kappa_{\rm subhalo} = \mathsf{R}\,\kappa_{\rm subhalo}$. (b) From the non-linearity: $g = \gamma/(1-\kappa)$ couples the subhalo to the main halo, and the NPE likelihoods are not exactly Gaussian, so the noise does not cancel perfectly. (c) A real cluster gives one map; there is no noise-matched twin without the subhalo. The difference map shows how visible the subhalo is *through the method*: its position, size and amplitude after the prior's smoothing. The significance a real map could claim comes from G4. (d) The rows of R near the subhalo are broad and low (F4(d)): the map shrinks and spreads it to about 40% of its peak.

G4 [Py] The significance one map can claim

Given. For each of K independent realizations of the cluster (new galaxies, positions and noise), we average the reconstructed κ over a 0.5 Mpc disk at the subhalo position: $a_{\mathrm{sub},r}$ for the main + sub map, $a_{\mathrm{main},r}$ for the main-only map. The single-map significance is $z_r = (a_{\mathrm{sub},r} - \overline{a_{\mathrm{main}}})/\mathrm{sd}(a_{\mathrm{main}})$, with the mean and sd over realizations. An aperture average of a mass map is a standard detection statistic (S96, Sec. 2.2 (Aperture measures), Eqs. (9) and (14)–(16), and Sec. 3 (Signal-to-noise statistics), Eq. (24); B&S01, Sec. 5.3.1, Eqs. (5.20)–(5.30)); the empirical-null z is *derived here*.

You'll need from before: B1, G3.

(a) Show that subtracting the same fixed map (for example the true main-halo κ) from every map leaves every z_r unchanged.

(b) Why is z_r computed against the other realizations' main-only maps, and not against realization r's own main-only map?

(c) Our null is built from maps of the *true* main halo. For a real cluster it would come from simulations of a *fitted* main halo. Why does that make our z an upper bound?

(d) With our numbers, $\overline{a_{\mathrm{main}}} = 0.0072$, sd $= 0.0038$ and $\overline{a_{\mathrm{sub}}} = 0.0156$, compute the z of the average realization.

(e) [Py] In the 1D toy of E5, generate 50 realizations with and without a narrow bump, compute z_r, and estimate the fraction with $z > 2$.

Solution. (a) Subtracting a fixed map subtracts the same constant c (its aperture average) from every $a_{\mathrm{sub},r}$ and every $a_{\mathrm{main},r}$. The difference in the numerator and the sd are unchanged. (b) Realization r's own main-only map shares its noise, which would cancel (G3), and a real observation has no such twin. The other realizations represent "what that spot looks like in a cluster without a subhalo", across noise and galaxy layouts. (c) Uncertainty in the fitted mass, concentration and centre adds scatter to the null, widening its sd and lowering z. (d) $(0.0156 - 0.0072)/0.0038 = 2.2$ (the median over realizations is 2.4). (e) The notebook (one seed) gives median $z \approx 1.9$, with 48% of realizations above $z = 2$: a toy version of our 2.4 and 60%. It also confirms (a): subtracting the truth changes no z.

G5 Reading the ensemble numbers

Given. Our results over 50 independent NPE realizations (`repo_reports/PAPER_DRAFT_HANDOFF.md`): one realization constrains m to $\approx \pm 0.3$; the pooled ensemble gives $m = +0.04 \pm 0.04$ (polar g_t); the pull spread is 0.84; the likelihood map's median z is 2.4 (KS: 1.1); and the best possible matched-filter S/N (C5(d), with full precision matrices) has median 3.7 over realizations. Statistical errors average down as $1/\sqrt{K}$ (A2).

You'll need from before: A2, B1, C5.

(a) Check $0.3/\sqrt{50}$ against the ensemble error. How many realizations would ± 0.01 need? Why doesn't the cluster test aim for that?

(b) What fraction of the matched-filter S/N does the likelihood map achieve, and what fraction does KS achieve? Name two reasons the map falls short of the matched filter.

(c) With 4 times the source density (Xu sample 1), roughly what median z would you expect, if $z \propto \sqrt{n}$?

Solution. (a) $0.3/\sqrt{50} = 0.042$, consistent. ± 0.01 needs $(0.3/0.01)^2 = 900$ realizations, because one cluster's cells span only ≈ 0.04 in shear and the slope is poorly pinned. The test asks a different question: no extra bias from shear gradients at the few-percent level. Percent-level m comes from the constant-shear stacks. (b) $2.4/3.7 = 65\%$; KS $1.1/3.7 = 30\%$. The matched filter knows the subhalo's exact profile and position, while the map assumes neither. The map's prior also shrinks the subhalo and the null includes galaxy-layout variance. (c) $\approx 2.4 \times 2 = 4.8$.

G6 Why the map settings were chosen on synthetic data

Given. We tried 27 likelihood-map settings and 4 KS settings. The rule "highest contrast with a false-peak rate ≤ 0.10" was fixed before the NPE maps were looked at, and applied to synthetic data only. For a standard normal, $P(z > 2) = 0.0228$. *Derived here.*

You'll need from before: B1, G4.

(a) If instead we had tried all 31 settings on the real map and reported the best z, and the settings were independent, what is the chance that at least one gives $z > 2$ when there is no subhalo?

(b) Real settings are correlated. Does that make the problem better or worse than (a)?

(c) Why does choosing on synthetic data, with the rule fixed in advance, avoid the problem?

Solution. (a) $1 - (1 - 0.0228)^{31} = 0.51$: a coin flip. (b) Better, since correlated settings behave like fewer independent tries, but the bias stays positive. (c) The choice never sees the data it is judged on, so the reported z is a single, pre-registered test.

Sources

Every section and equation number cited in the Given boxes was read in the source text (or, for scanned and badly extracted pages, in rendered page images); `references.md` quotes the line each citation points to. Equation numbers follow the arXiv versions where those were used.

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